REP-028 · PLAN-012 built: market rebalance, optional tweaks and the cosy home screen

Game	Fruit Market, rules version 3 (branch `workflow/plan-012`)
Report date	2026-10-27 10:35 (Asia/Kuala_Lumpur)
Type	Implementation and verification of PLAN-012 (approved: "yes approvrd together with the optional changes and the new layout from the mockup")
Builds on	REP-027 (balance proposal)
Approval status	Approved (PLAN-012). Publishing the live game still needs its own approval

How to read the numbers.

- **[Code]** marks a fact about the code.
- **[Sim]** marks a computer-player result: 600 games per test, about ±4 points, on seeds 100001–100600, which were never used for tuning.
- **[Hyp]** marks a guess about people. Only the playtest can confirm it.

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1. What was built

Item	Change	Where
E1	Customers start at 60% of the town's buying power (was 40%), growing +45%. Farmers bring 80% + 20% growth (was 100% + 50%)	`MARKET` in `src/engine/data.ts`; `demandScale`, `supplyScale` in `src/engine/game.ts`
E2	The town is sized for 4 fruits per shop. Handcart and Big Basket help only their owner	`marketCapacity()` in `game.ts`; Handcart text
E3	4-shop towns have 15% more customers	`MARKET.fourPlayerDemand`
E4	Apples cost $3 / $2 / $1 (were $4 / $3 / $1)	`FRUIT.apple`
E5	`RULES_VERSION` 3; daily boards keyed by rules version (`daily:<date>:r3`), so release-day scores never mix	`src/net/protocol.ts`, `dailyBoardKey()` in `server/meta.ts`, `server/worker.ts`
O1	Haggler wins ties against richer shops before freshness is compared. Local Favourite tip $4 (was $3). Nest Egg starts with $15 (was $16)	`hagglerEdge()` in `game.ts`; `TRAITS`, `NEST_EGG_BONUS`, `LOCAL_FAVOURITE_TIP`
O2	Green mango costs $3 (was $2)	`FRUIT.mango`
O3	**Tomorrow's lorry**: tomorrow's farmer card is drawn a day early and shown to everyone as plenty or few per fruit. Computer players use it to hold scarce fruit	`nextFarmer` in `game.ts`; `NextLorry` in `src/ui/table.tsx` (market and morning card); `src/engine/ai.ts`
O5	**Juice Press**: each older fruit sold (okay or not so) earns $1 extra as juice; the nightly juicing of spoiling fruit stays. The sale pop-up includes it	`JUICE_TIP`; `resolve()` in `game.ts`; `src/ui/game/animator.tsx`

Item	Change	Where
O5	**Cool Room**: unchanged (see section 3)	–
O4	**Long (14-day) games**: unchanged (see section 3)	–
O6	**Market League** stages retuned (section 2d)	`LEAGUE` in `src/ui/progress.ts`
L1	**Home screen in the cosy pixel layout** from the mockup: harbour scene with bobbing boats, bunting, the "Fruit Market" sign with Foxy Rosa peeking out, fruit on the pier, then New game, Play a friend / Quick match, Daily challenge and Market League cards, and How to play / Trophies / Settings. Motion stops when reduced motion is on	`src/ui/App.tsx` , `src/ui/styles.css`
Text	Rules page (fruit table, the town grows, tomorrow's lorry, ties, Handcart, Juice Press, traits, balance notes). The wrong apple price-cap example is fixed. How-to mentions the lorry. Sale explanations name the Haggler rule	`RULES.md` , `src/ui/HowTo.tsx` , `src/engine/story.ts`

Saves and online play.

- A game in progress from the old version shows the existing "older version of the rules" notice and is cleared (`src/ui/store.ts` , unchanged).
- Old app versions are refused by online rooms and the daily board with the existing "please update" message.

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2. Evidence

2a. The game code plays exactly like the tested package

[Sim] Production engine against the lab with the package switched on: **450 games** (2–4 players, 8/10/14 days, mixed levels). **0 differences in final cash.**

2b. Strategy matchups, 3 players (fair share 33%)

One plan against two Hard computers.

Plan	Before (rules 2)	After (rules 3)
Undercut	43	27
Dump old stock (Salvage)	51	36
Premium prices	13	28
Reserve (keep cash)	30	36
Ripen green fruit	27	34
Read rivals (Adaptive)	36	37
Turnover	18	23
Events	32	29
Spread	**38 points**	**14 points**

In 4 players (fair share 25%): Undercut 45 → 20, Salvage 54 → 28, Premium 6 → 20.

Four plans at one 4-player table:

	Turnover	Reserve	Ripen	Events
Before	17	19	29	34
After	16	32	33	20

2c. Pacing and the success criteria from REP-027

#	Criterion	Before	After	Result
S1	No plan above 42% (3p) / 31% (4p)	51% / 54%	37% / 28%	**PASS**
S2	At least 4 of 8 plans within ±7 of fair (3p)	3	7	**PASS**
S3	Hard beats Normal in 2 players at least 65% of the time	80%	**60%**	**FAIL** (see 3)
S4	Comeback from last at half-time, 3p at least 20% / 4p at least 15%	15.3% / 8.2%	25.8% / 19.2%	**PASS**
S5	4-player cash at least 60% of 2-player cash	43% ($56 / $130)	67% ($100 / $150)	**PASS**
S6	Price-war sales at most 10%	11.4% / 16.3%	5.8% / 6.7%	**PASS**
S7	Long mode: no plan above 42% at a 4-plan table	–	42% (Reserve); Reserve alone wins 50% against Hard	**FAIL** (see 3)
S8	Human enjoyment	–	–	**NOT RUN** (playtest)

Other measures, 3 players:

- Blowouts: 45.5% → 36.3%.
- Mid-game leader wins: 56.8% → 43.5%.
- Lead changes per game: 3.6 → 4.3.
- Average cash: 2 players $130 → $150, 3 players $80 → $108, 4 players $56 → $100.

Quick (8-day) games are the most even length. At the 4-plan table the plans score 23 / 28 / 25 / 24. Salvage reaches 40% against Hard computers, still under the 42% limit.

2d. Traits, upgrades and the league

Traits (2,000 games per player count):

- 3 players: from 29.9% (Charmer) to 37.7% (Bargain Hunter), an 8-point spread (was 10).
- 4 players: from 23.3% to 28.5%, a 5-point spread (was 11.5).

Upgrades. A Hard computer building only one upgrade, 3 players. Building nothing wins 10.8%.

Upgrade	Win %
Signboard	20.7
Cool Room	15.8
Handcart	14.7
Shop Bell	13.5
Juice Press	12.8 (was below nothing: 9.7)
Bigger Shelf	12.2

Every upgrade is now worth more than none.

Market League (300 games per stage, a computer standing in for you):

Stand-in	Village	Harbour	Old Town	Spice Lane	Grand Bazaar	Royal Market
Normal, rules 2 and old league	65	67	54	38	11	5
Normal, rules 3 and new league	64	64	52	41	33	20
Hard, rules 3 and new league	76	76	64	58	48	35

New stages:

1. Village Square: 1 Easy rival, 8 days.
2. Harbour Steps: 1 Easy rival, 8 days.
3. Old Town: 1 Normal rival.
4. Spice Lane: 2 Easy rivals.
5. Grand Bazaar: 2 Normal rivals.
6. Royal Market: 1 Hard and 2 Normal rivals.

Stages 3–6 are 10 days. The old final stage (three Hard rivals) was a wall.

2e. Checks

Check	Result	Evidence
`npm run verify` (records, lint, types, 81 unit tests, build)	PASS	exit 0
`npm run verify:full` (browser tests, online leak check, reconnect, quick match, four players)	PASS	exit 0; see section 5
Home screen at 390×844, 360×640 and 1280×800	PASS	no sideways scroll, no page errors; screenshots checked
Tomorrow's lorry visible on the morning card and under the market	PASS	screenshot at 390×844

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3. What did not work, and what I did instead

- **Long games (O4).** I tried 10 different levers in the lab:
    - demand growth, demand start, supply start, supply growth;

    - when the second walk-in arrives;
    - the Cool Room price;
    - and others.

  None stopped patient plans (Reserve and Ripen) winning most 14-day games; Reserve stays at 39–52%. Long mode therefore keeps the standard rules. [Hyp] Patience winning a long game may feel right to players. The playtest should check it.

- **Cool Room (O5).**

    - A cheaper Cool Room ($4–5) adds little.
    - A Cool Room that stops ageing completely makes the Reserve plan too strong (45–49%).
    - Under the new market the current Cool Room is already worth buying (15.8%, against 10.8% for nothing), so it stays unchanged.

- **Juice Press (O5).** Four juice designs were tested. Juicing any leftover fruit was worse, because it turns good fruit into juice. The +$1 "juice tip" on older fruit was the only design that helped without boosting any plan.

- **Skill gap (S3).** Hard beats Normal 60% of the time (target 65%). Most of the drop comes from green mango at $3: without it the figure is about 70%, but Ripen and Turnover weaken.

    - Mango is on the preview as approved.
    - Changing it back to $2 is a one-line change. Decide after playtesting.
    - [Hyp] A smaller gap makes the game friendlier to new players and less punishing against Hard.

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4. Risks

- Fewer spare fruits on Day 1: the market looks sparser. [Hyp] This may feel tight to new players.
- The $200 "Fruit Tycoon" and $40 "Market Day" achievements become easier (thresholds kept, as recommended).
- League progress is kept by stage number, so a player midway through keeps their stage under the new line-up.

5. Browser tests (`npm run verify:full`, exit 0)

- **Solo game:** results screen reached, no page errors.
- **Performance budget:** BUDGET OK.
- **Online match:**
    - 40 frames checked during pricing, 0 leaks of secret prices;
    - a refresh rejoined on Day 2;
    - both phones showed the same final cash;
    - rematch started.
- **Abandoned match:** a computer took over and the other player finished alone.

- **Quick match:** both players reached the game.
- **Four players:** lobby counts OK, a 5th person refused, all 4 in the game.
- **Reconnect stress:** 5 cycles, 0 failed (median 255 ms after a reload, 538 ms after a drop).
- **Old apps:** turned away by online rooms (`tests/room.test.ts`) and by the daily board (`tests/meta.test.ts`).

Lab files (engine copy, sweeps 7–11, parity check against the game code) are archived in `docs/workflow/reports/img/REP-028-lab.tar.gz` .